

International Islamic University Chittagong

Department of Computer Science and Engineering
Course: CSE-3529 | System Analysis and Design

■ COMPLETE EXAM ANSWER GUIDE

Midterm Exams: Autumn 2023 | Spring 2024 | Spring 2025 | Autumn 2025

All Questions Solved with Memory Tricks ✓ Math Solved Step-by-Step ✓

PART A — CORE CONCEPT ANSWERS

(These answers apply across multiple exam sittings)

■ **Q1a [Spring 2024, Autumn 2023]: Define system. Discuss the stages in building a system. Who is a system analyst? Explain role, responsibilities, and qualities.**

→ **SYSTEM:**

A **system** is a set of interrelated components working together to achieve a common goal. It takes **Input** → **Process** → **Output** and includes feedback.

Examples: Payroll system, Hospital management system, Banking system.

■ **TRICK:** Remember: $IPO + Feedback = System$ (Input → Process → Output → Feedback)

→ **STAGES IN BUILDING A SYSTEM (SDLC Phases):**

- **1. Planning:** Define objectives, scope, feasibility
- **2. Analysis:** Study existing system, gather requirements
- **3. Design:** Create blueprints — logical & physical design
- **4. Implementation:** Coding, testing, installation
- **5. Maintenance:** Bug fixes, updates, improvements

■ **TRICK:** Mnemonic: **PADIM** → Planning, Analysis, Design, Implementation, Maintenance

→ **SYSTEM ANALYST:**

A system analyst bridges the gap between users and technical team. They study, design, and implement information systems.

Roles: Requirements gathering, feasibility check, system modeling, user liaison.

Responsibilities: Define system requirements, prepare DFDs/ERDs, coordinate with programmers, test system.

Qualities: Good communication, analytical thinking, problem-solving, technical knowledge, teamwork.

■ **TRICK:** Think of analyst as a **TRANSLATOR** — translates user needs into technical specs!

■ **Q2a [All Exams]: What is SDLC? Briefly describe the different phases of SDLC with figure.**

SDLC (Software Development Life Cycle) is a structured process for planning, creating, testing, and deploying software. It ensures quality and reduces risk.

Phases of SDLC:

- **1. Planning:** Identify problem, define scope, estimate resources & cost, feasibility study.
- **2. System Analysis:** Study current system, gather user requirements, create requirement specification.

- **3. System Design:** Logical design (data flow, ER diagrams), Physical design (hardware, software specs).
- **4. Implementation:** Write code, unit testing, integration testing, user acceptance testing.
- **5. Maintenance:** Corrective (fix bugs), Adaptive (new OS), Perfective (new features).

Phase	Key Activity	Output
Planning	Feasibility study	Project plan
Analysis	Req. gathering	SRS document
Design	System blueprints	Design specs
Implementation	Coding & testing	Working software
Maintenance	Updates & fixes	Updated system

■ **TRICK:** SDLC = Plan → Analyze → Design → Build → Maintain. Like building a house!

■ **Q2b [Autumn 2025]: Briefly describe Waterfall model, Spiral model and Agile model. Which is better? Why?**

→ **WATERFALL MODEL:**

Linear sequential model. Each phase must complete before the next begins. No going back.

- Phases: Requirements → Design → Coding → Testing → Maintenance
- **Advantage:** Simple, easy to manage, good documentation.
- **Disadvantage:** Inflexible — changes are costly. Not suitable for complex projects.

■ **TRICK:** Waterfall = stairs going DOWN. You can't go back up easily!

→ **SPIRAL MODEL:**

Combines Waterfall + Iterative + Risk Analysis. Project goes through repeated cycles (spirals).

- Each spiral: Planning → Risk Analysis → Engineering → Evaluation
- **Advantage:** Best for large, high-risk projects. Risk handled at each phase.
- **Disadvantage:** Expensive, complex, requires expert risk assessment.

■ **TRICK:** Spiral = a coil spring — goes round and round but forward each time!

→ **AGILE MODEL:**

Iterative development in small chunks called **sprints** (2–4 weeks). Customer involved throughout.

- Phases within each sprint: Plan → Design → Build → Test → Review
- **Advantage:** Very flexible, fast delivery, customer-friendly, easy to adapt to changes.
- **Disadvantage:** Hard to predict timeline/cost, needs active customer participation.

■ **TRICK:** Agile = moving fast & flexible — like a sports car, not a truck!

→ **WHICH IS BETTER?**

Agile is generally better for modern software development because:

- ✓ Requirements change frequently in real world
- ✓ Early delivery of working software
- ✓ Continuous customer feedback improves quality
- ✓ Teams respond quickly to market changes

However, **Waterfall** suits projects with fixed, clear requirements (e.g., government contracts). **Spiral** suits high-risk, large-scale systems (e.g., NASA software).

■ **Q2a [Spring 2024]: Discuss RAD (Rapid Application Development) and JAD (Joint Application Development) with advantages and disadvantages.**

→ **RAD (Rapid Application Development):**

A model focused on **rapid prototyping** over planning. Uses small, specialized teams and tools.

- Steps: Requirements → Prototype → Build → Test (Repeated fast)
- **Advantages:** Fast development, early user feedback, flexible, reduced cost
- **Disadvantages:** Needs skilled team, not for large projects, poor documentation

■ **TRICK:** RAD = Quick prototype first, perfect later. Like drafting with pencil before final pen!

→ JAD (Joint Application Development):

A structured workshop-based technique where users and developers **jointly** define requirements.

- JAD sessions involve: Users, Analysts, Managers, Facilitator
- **Advantages:** Better requirement accuracy, user buy-in, saves time vs individual interviews
- **Disadvantages:** Expensive (all stakeholders in one room), needs skilled facilitator

■ **TRICK:** JAD = All in one room, decide together. Like a team huddle before the game!

■ Q1a [Autumn 2025]: Discuss system analysis. Discuss the stages in building an improved system with diagram.

System analysis is the process of studying a current system to understand problems and determine how a new or improved system can solve them. It answers: **WHAT does the system do?**

Stages in Building an Improved System:

- **Stage 1 — Problem Recognition:** Identify that current system has issues (slow, errors, inefficient)
- **Stage 2 — Feasibility Study:** Check if solution is technically, economically, and operationally possible
- **Stage 3 — Requirement Analysis:** Gather detailed user needs through interviews, surveys, observation
- **Stage 4 — System Design:** Create logical (DFD, ERD) and physical design (hardware/software)
- **Stage 5 — Implementation:** Develop, test, and deploy the new system
- **Stage 6 — Evaluation & Maintenance:** Monitor performance and update regularly

■ **TRICK:** PARFIE: Problem → Analysis → Requirements → Feasibility → Implementation → Evaluation

■ Q1b [Autumn 2025] / Q1a [Spring 2025]: Discuss quality/characteristics of information. OR Define information and key qualities that make it valuable.

Information is processed data that is meaningful and useful for decision-making.

Key Qualities of Good Information (ACTURAC):

Quality	Meaning	Example
Accurate	Free from errors	Correct student marks
Complete	All needed details present	Full customer record
Timely	Available when needed	Today's stock prices
Relevant	Useful for the decision	Sales data for manager
Understandable	Clear and readable	Simple charts
Reliable	Trustworthy source	Official govt data
Accessible	Easy to retrieve	Online database

■ **TRICK:** Mnemonic: ACTURAR = Accurate, Complete, Timely, Understandable, Relevant, Accessible, Reliable

■ Q2a [Spring 2025] / Q1b [Spring 2024]: What is a system? Discuss different types of information systems and their roles in organizations.

A system is a collection of interrelated components that work together toward a common goal.

Types of Information Systems:

- **1. TPS (Transaction Processing System):** Handles daily business transactions. Ex: ATM, point-of-sale
- **2. MIS (Management Information System):** Provides reports for middle managers. Ex: Monthly sales report
- **3. DSS (Decision Support System):** Helps managers make semi-structured decisions. Ex: Loan approval
- **4. ESS/EIS (Executive Support System):** For top executives — strategic decisions. Ex: Market trend graphs
- **5. OAS (Office Automation System):** Supports office work. Ex: Email, word processing

■ **TRICK:** Remember levels: TPS (bottom/daily) → MIS → DSS → ESS (top/strategy). Like floors in a building!

■ Q1a [Autumn 2023] / Q1b [Spring 2024]: Classify types of information as Strategic, Tactical, and Operational. (Foodpanda / TechFiesta scenario)

Level	Who Uses It	Purpose	Example (Foodpanda)
Strategic	Top executives/CEO	Long-term decisions	Market expansion plan
Tactical	Middle managers	Medium-term planning	Weekly promotion plans
Operational	Front-line staff	Day-to-day operations	Order tracking, delivery status

Foodpanda scenario answers:

- i. In-app pop-up notifications (new features) → **Operational** — daily user interaction
- ii. Top 10 meals / top-rated restaurants → **Tactical** — weekly planning & promotion
- iii. Trending feature (calorie count) → **Strategic** — long-term app positioning

TechFiesta Mega Sale scenario (Spring 2024 Q1b):

- **Strategic:** Decision to launch TechFiesta promotion to capture market share (CEO level)
- **Tactical:** Planning 30% discount duration (Jan 1–15), targeting electronics category (manager level)
- **Operational:** Updating prices in system, processing orders, staff scheduling during sale (operational staff)

■ **TRICK:** Think of it as a company PYRAMID: Operational = base (many, daily), Tactical = middle, Strategic = top (few, long-term)

■ Q3a [Autumn 2025, Spring 2025]: Discuss feasibility study — why it is done, and its areas. (Significance of feasibility analysis in project management)

A **feasibility study** determines whether a proposed system is worth developing. It is done before spending major resources on development.

Why it is done:

- ✓ Avoids wasting money on impractical projects
- ✓ Identifies risks early
- ✓ Helps compare alternative solutions
- ✓ Provides basis for project approval

Three Major Areas of Feasibility (TOES):

- **1. Technical Feasibility:** Does the organization have the technology and expertise to build it? (hardware, software, skills)
- **2. Economic/Financial Feasibility:** Is the benefit greater than cost? Includes cost-benefit analysis, ROI, BEP, NPV

- **3. Operational Feasibility:** Will users accept and use the system? Will it fit the organization's workflow?
- (Also: **Schedule Feasibility** — Can it be done in time? **Legal Feasibility** — Are there legal issues?)

■ **TRICK: Mnemonic: TELOS** = Technical, Economic, Legal, Operational, Schedule

Feasibility study is done in the **Planning Phase** of SDLC — right at the beginning, before any design.

■ **Q3b [Spring 2025]: Explain the role of SWOT analysis in strategic planning for IT projects. How does SWOT help IT managers?**

SWOT Analysis is a strategic planning tool that identifies internal and external factors affecting a project.

Factor	Type	Meaning	IT Example
Strengths	Internal (+)	Advantages you have	Expert dev team, modern tech
Weaknesses	Internal (-)	Areas to improve	Old legacy system, budget limit
Opportunities	External (+)	Favorable conditions	Growing digital market
Threats	External (-)	Risks from outside	Competitor software, cyber attacks

How SWOT helps IT managers:

- ✓ Identifies gaps in skill/technology before project starts
- ✓ Helps prioritize resources to Strengths
- ✓ Prepares contingency plans for Threats
- ✓ Helps align IT strategy with business goals

■ **TRICK: SWOT** = Know yourself + Know your environment. Like a chess player analyzing the board!

■ **Q3a [Spring 2025]: Discuss both internal and external factors that affect a system project.**

→ **INTERNAL FACTORS (Inside the organization):**

- **Management support:** Without top-level support, project fails
- **Budget constraints:** Limited funds delay or downgrade project
- **Staff skills:** Unskilled team can't deliver quality system
- **Organizational culture:** Resistance to change slows adoption
- **Existing systems:** Old legacy systems may be incompatible

→ **EXTERNAL FACTORS (Outside the organization):**

- **Government regulations:** Data protection laws, compliance requirements
- **Competition:** Rivals launching new features forces faster delivery
- **Technology change:** New tools/platforms may render design obsolete
- **Economic conditions:** Recession reduces IT budget approval
- **User/market demand:** Changing user expectations require system updates

■ **TRICK: Internal** = things YOU control. **External** = things the WORLD throws at you!

■ **Q3a [Spring 2024]: Define project management. Describe the execution phase of project management.**

Project Management is the application of knowledge, skills, tools, and techniques to plan and control project activities to meet specific goals within a defined scope, time, and budget.

Execution Phase: This is where the actual work happens — the project plan is put into action.

- **1. Direct & manage work:** Team starts building/coding per the plan
- **2. Quality assurance:** Checking outputs meet standards throughout development
- **3. Team management:** Assign tasks, resolve conflicts, motivate team members

- **4. Stakeholder communication:** Regular status updates, progress reports
- **5. Procurement:** Purchase tools, equipment, hire vendors if needed
- **6. Risk management:** Monitor and respond to risks as they occur

Key output of Execution Phase: Working deliverables, performance reports, change requests

■ **TRICK:** Execution Phase = 'Do the work'. Plan was the MAP, Execution is the JOURNEY!

■ Q2b [Autumn 2023]: Distinguish between Methodologies, Models, Tools and Techniques with examples.

Term	Definition	Example
Methodology	A structured set of guidelines/procedures for a process	Scrum, Agile, RUP, DevOps
Model	A simplified representation of a system	ER Diagrams, DFD, UML
Tool	Software/hardware used to support a process	MS Visio, R, Actix, SQL
Technique	A specific way to perform a task within a process	Interviewing, observation, prototyping

■ **TRICK:** Analogy: Methodology=Recipe book, Model=Blueprint, Tool=Kitchen equipment, Technique=Chopping skill!

■ Q3b [Autumn 2023] / OR questions [Multiple exams]: Describe cost-benefit analysis and steps to conduct economic feasibility analysis.

Cost-Benefit Analysis (CBA) compares the expected costs of a system against the expected benefits to determine if the project is economically worthwhile.

Key Metrics:

- **ROI (Return on Investment):** Shows how profitable the investment is (as %)
- **BEP (Break-Even Point):** Year when cumulative benefits equal cumulative costs
- **NPV (Net Present Value):** Present value of future benefits minus costs

Steps to Conduct Economic Feasibility Analysis:

- **Step 1:** Identify all costs — development cost, hardware, software, training, maintenance
- **Step 2:** Identify all benefits — tangible (increased revenue, cost savings) and intangible (better decisions)
- **Step 3:** Estimate cash flows for each year of the project
- **Step 4:** Calculate ROI, BEP, and NPV
- **Step 5:** Compare results with organization's financial goals — recommend Go/No-Go

■ **TRICK:** CBA = Is it worth it? Benefits > Cost → GO! Benefits < Cost → NO GO!

■ Q3b [Autumn 2023 OR]: Define: Deliverables, Break-Even Point, Net Present Value, Recurring Cost, Return on Investment.

i) Deliverables: Tangible or intangible products/outputs produced at the end of a project phase. Ex: SRS document, database design, final software.

ii) Break-Even Point (BEP): The year/point when total cumulative benefits equal total cumulative costs. After BEP, the project becomes profitable.

BEP = Year when Cumulative Net Cash Flow crosses 0 (from negative to positive)

iii) Net Present Value (NPV): The current value of all future net benefits, adjusted for the time value of money. If NPV > 0, project is profitable.

NPV = Sum of [Net Benefit / (1 + r)ⁿ] for each year n, where r = discount rate

iv) Recurring Cost: Costs that repeat annually throughout the project's life. Ex: maintenance fees, staff salaries, software licenses, server hosting.

v) Return on Investment (ROI): Percentage measure of profit relative to investment cost.

$$\text{ROI} = (\text{Total Net Benefits} / \text{Total Cost}) \times 100\%$$

■ *TRICK: ROI tells HOW MUCH you gain. BEP tells WHEN you start gaining. NPV tells TODAY'S VALUE of future gains!*

PART B — MATHEMATICS: FULLY SOLVED

ROI, BEP, and NPV calculations from all exam papers

■■■ MATH 1 — Autumn 2025 (Q3b) ■■■

■ Q3b [Autumn 2025]: From the cash flow table, calculate (i) Return on Investment and (ii) Break-Even Point.

Given Data from Table:

Description	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits	0	45,000	50,000	57,000	1,52,000
Total Cost	1,00,000	10,000	12,000	16,000	1,38,000
Net Benefits	(1,00,000)	35,000	38,000	41,000	14,000
Cumulative NCF	(1,00,000)	(65,000)	(27,000)	14,000	—

(i) ROI — Return on Investment:

Formula: $\text{ROI} = (\text{Total Net Benefits} / \text{Total Development Cost}) \times 100\%$

Total Net Benefits (from table) = 14,000

Total Cost (development/initial cost) = 1,00,000 (Year 0)

$\text{ROI} = (14,000 / 1,00,000) \times 100\%$

$\text{ROI} = 0.14 \times 100\%$

ROI = 14%

■ This means for every 100 taka invested, the organization gains 14 taka profit over 3 years.

(ii) BEP — Break-Even Point:

Formula: Look at Cumulative Net Cash Flow — BEP is when it crosses 0.

Year 0: Cumulative NCF = -1,00,000 (negative)

Year 1: Cumulative NCF = -1,00,000 + 35,000 = -65,000 (still negative)

Year 2: Cumulative NCF = -65,000 + 38,000 = -27,000 (still negative)

Year 3: Cumulative NCF = -27,000 + 41,000 = +14,000 (positive! ✓)

→ BEP occurs between Year 2 and Year 3.

Exact BEP = Year 2 + (Amount still to recover at end of Year 2 / Net Benefit in Year 3)

Exact BEP = 2 + (27,000 / 41,000)

Exact BEP = 2 + 0.658

Exact BEP ≈ 2.66 years (i.e., approximately 2 years and 8 months)

■ The project recovers its full investment in about 2 years and 8 months.

■ **TRICK: BEP Trick:** Find the year where Cumulative NCF changes from Negative to Positive. The BEP is in THAT year!

■■■ MATH 2 — Spring 2024 (Q3b) ■■■

■ Q3b [Spring 2024]: For an IT project, Year 0,1,2,3 costs are 50,000 / 8,000 / 10,000 / 14,000 and benefits are 0 / 25,000 / 30,000 / 48,000. Find (i) ROI and (ii) BEP.

Step 1: Build the Cash Flow Table

Year	Benefits	Cost	Net Benefit (B - C)	Cumulative NCF
Year 0	0	50,000	-50,000	-50,000
Year 1	25,000	8,000	+17,000	-33,000
Year 2	30,000	10,000	+20,000	-13,000
Year 3	48,000	14,000	+34,000	+21,000
TOTAL	1,03,000	82,000	+21,000	—

(i) ROI:

$ROI = (\text{Total Net Benefits} / \text{Total Development Cost}) \times 100\%$

Total Net Benefits = 21,000

Total Development Cost = 50,000 (Year 0 investment)

$ROI = (21,000 / 50,000) \times 100\%$

$ROI = 0.42 \times 100\%$

ROI = 42%

■ Very good ROI! For every 100 taka invested, 42 taka profit is earned over 3 years.

(ii) BEP:

Tracking Cumulative NCF:

Year 0: -50,000

Year 1: -50,000 + 17,000 = -33,000

Year 2: -33,000 + 20,000 = -13,000

Year 3: -13,000 + 34,000 = +21,000 ← crosses zero here!

$BEP = 2 + (13,000 / 34,000)$

$BEP = 2 + 0.382$

BEP ≈ 2.38 years (approximately 2 years and 5 months)

■ Project becomes profitable in about 2 years and 5 months.

■ MATH FORMULA QUICK REFERENCE

Formula	Expression	What it means
ROI	$(\text{Total Net Benefits} / \text{Total Cost}) \times 100\%$	Profit % on investment
Net Benefit	Benefits - Costs (per year)	Yearly gain/loss
Cumulative NCF	Sum of all net benefits up to that year	Running total
BEP (simple)	Year when Cumulative NCF = 0	Payback year
BEP (exact)	$\text{Year } N + (\text{NCF at end of Year } N / \text{Net Benefit in Year } N+1)$	Precise payback
NPV	$\sum [\text{Net Benefit}_n / (1+r)^n]$ for n=0 to N	Present value of profits

PART C — SCENARIO & APPLIED QUESTIONS

■■■ Spring 2025 — Q1b Scenario ■■■

■ **Q1b [Spring 2025]:** A software team has high-risk factors and frequent requirement changes. Manager divides into smaller phases with risk assessment, prototyping, and client feedback after each. What software development model is being used?

Answer: SPIRAL MODEL

This is the Spiral Model because:

- ✓ Project divided into smaller phases (spirals) — matches the scenario
- ✓ Risk assessment done at each phase — key feature of Spiral
- ✓ Prototyping is used — Spiral uses evolutionary prototyping
- ✓ Client feedback after each phase — iterative nature of Spiral

How it addresses challenges:

- Risk management: Each spiral starts with risk analysis — risks are identified and mitigated early
- Iterative development: Builds on previous spiral — continuously refines the system
- Prototyping: Working prototype after each spiral — client can verify before full build

■ **TRICK: KEY:** *If question mentions RISK + PHASES + PROTOTYPING + REVIEW → Always SPIRAL MODEL!*

■ **Q2a [Autumn 2023 / Spring 2025 OR]:** Which SDLC model would you adopt for developing an e-commerce platform with small requirements and no uncertain requirements? Justify.

Answer: WATERFALL MODEL

Justification:

- ✓ Requirements are small and fixed — no changes expected (perfect for Waterfall)
- ✓ Linear sequential approach works well when scope is clear
- ✓ Easy to manage with clear documentation at each phase
- ✓ No need for prototyping or iterations — saves time and cost

Advantages: Simple structure, well-defined phases, excellent documentation, easy to test

Disadvantages: If requirements change later, costly to go back; late testing phase

■ **TRICK:** *Small + Fixed requirements = WATERFALL. Large + Changing requirements = AGILE or SPIRAL!*

■ **Q2a [Autumn 2023 OR]:** Online learning platform for instructors/students with regular feedback. Which SDLC model?

Answer: AGILE MODEL (Specifically SCRUM)

Justification:

- ✓ Platform needs regular updates based on educator/student feedback — Agile excels here
- ✓ Features like quizzes, progress tracking, forums can be delivered in sprints
- ✓ Continuous feedback loop between educators and developers — Agile's core strength
- ✓ Market needs change — Agile adapts quickly

Advantages: Flexible, fast delivery, customer-centered, easy iteration

Disadvantages: Hard to predict total timeline, requires constant customer involvement

■ **TRICK:** *Continuous feedback + changing features + evolving content = AGILE MODEL!*

■ Q2b [Spring 2024]: (i) Accounting software with detailed upfront requirements, short delivery time. (ii) Social networking platform adapting to market quickly with iterative features. Which SDLC for each?

i) Accounting Software → WATERFALL MODEL

- Client provided detailed requirements upfront — Waterfall is ideal (fixed requirements)
- Short delivery time with clear scope — no need for iterations
- Structured testing phase ensures accurate accounting functions

ii) Social Networking Platform → AGILE MODEL

- Needs to adapt to changing market quickly — Agile's strength
- Release features iteratively and gather user feedback — core of Agile
- Evolving requirements can be handled with sprint-based development

■ *TRICK: Upfront clear requirements + short time = WATERFALL. Changing needs + iterations = AGILE!*

■ Q1b [Autumn 2025 OR]: Discuss the qualities of a system analyst and perspectives of stakeholders on an information system.

Qualities of a System Analyst:

- ✓ Analytical thinking — ability to break down complex problems
- ✓ Communication skills — talks to users, managers, and developers effectively
- ✓ Technical knowledge — understands hardware, software, databases
- ✓ Problem-solving — finds practical solutions to system issues
- ✓ Flexibility — adapts to changing requirements and technology
- ✓ Interpersonal skills — builds trust with stakeholders

Stakeholder Perspectives on Information System:

Stakeholder	Their Perspective
Top Management (CEO/Director)	Focus on ROI, strategic goals, competitive advantage
Middle Managers	Need accurate reports for decision-making, ease of use
End Users	Want simple UI, speed, reliability, minimal training
IT Department	Focus on maintenance, security, scalability, compatibility
Customers/Clients	Want fast service, privacy, 24/7 availability

■ *TRICK: Each stakeholder sees the system through their OWN lens — analyst must balance ALL perspectives!*

PART D — QUICK REVISION CHEAT SHEET

Topic	Key Trick / Mnemonic	Answer in 1 Line
SDLC Phases	<i>PADIM</i>	Plan→Analyze→Design→Implement→Maintain
System	<i>IPO + Feedback</i>	Input→Process→Output with feedback loop
Waterfall	<i>Stairs going DOWN</i>	Linear, sequential, no going back
Agile	<i>Sports car — fast & flexible</i>	Sprint-based, customer-centric, adaptive
Spiral	<i>Coil spring — round but forward</i>	Risk-driven, iterative prototyping
RAD	<i>Draft with pencil first</i>	Fast prototyping, small teams, quick delivery
JAD	<i>Team huddle</i>	Workshop where users+devs define requirements together
Feasibility	<i>TELOS</i>	Technical, Economic, Legal, Operational, Schedule
SWOT	<i>Know yourself + Know world</i>	Strengths, Weaknesses, Opportunities, Threats
ROI Formula	<i>$(\text{Net Benefits} / \text{Cost}) \times 100$</i>	How profitable is the investment?
BEP	<i>When cumulative NCF = 0</i>	Year project starts making profit
Quality Info	<i>ACTURAR</i>	Accurate, Complete, Timely, Understandable, Relevant...
Info Types	<i>CEO-Manager-Staff pyramid</i>	Strategic→Tactical→Operational
TPS→ESS	<i>Bottom to top of company</i>	TPS (daily) → MIS → DSS → ESS (strategy)
System Analyst	<i>TRANSLATOR</i>	Bridges gap between users and technical team

Good luck in your exams, Ahsan! ■

CSE-3529 | International Islamic University Chittagong